

SHRI RAMDEOBABA COLLEGE OF ENGINEERING AND MANAGEMENT, NAGPUR



Department of Computer Science & Engineering

PROJECT SYNOPSIS

Session: 26-27

Semester: 7th

Programme: B. Tech in Computer Science & Engineering (Data Science) / B. Tech in Information Technology

Explainable AI-Based Red Blood Cell Morphology Classification and Clinical Report Generation Using EfficientNetV2

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Date:

[03 / 08 / 2026]

1. Background and Motivation

Red Blood Cell (RBC) morphology plays a crucial role in diagnosing various hematological disorders, including thalassemia, iron deficiency anemia, hereditary spherocytosis, liver disease, and other hemoglobin abnormalities. Microscopic examination of peripheral blood smear images allows hematologists to identify abnormalities in RBC shape, size, and color. However, manual examination is time-consuming, subjective, and requires experienced medical professionals, making it difficult to achieve consistent and efficient diagnosis in high-volume clinical settings.

Recent advancements in Artificial Intelligence (AI) and Computer Vision have enabled automated analysis of blood smear images using deep learning techniques. The selected base paper proposes an automated framework for RBC segmentation, overlapping cell separation, and classification while addressing the challenge of severe class imbalance. Although the proposed framework achieves promising classification performance, it primarily focuses on accurate cell identification and does not provide explainability, prediction confidence, or clinically interpretable outputs.

To overcome these limitations, this project extends the base research by integrating an EfficientNetV2-based transfer learning model with Explainable AI techniques such as Grad-CAM for transparent decision-making. The proposed system also generates confidence scores and an automated clinical report describing the predicted RBC morphology and its possible clinical significance. By combining accurate classification with explainability and clinical reporting, the proposed system aims to serve as an intelligent decision-support tool for hematologists and pathology laboratories.

2. Problem Statement

Automated RBC morphology classification has gained significant attention due to its potential to improve the efficiency and consistency of hematological diagnosis. Although recent research has successfully addressed challenges such as overlapping cell separation and class imbalance, most existing systems function as black-box classifiers and provide only the predicted class label.

The absence of visual explanations, confidence estimation, and automated clinical interpretation limits their usefulness in real-world medical applications. Therefore, there is a

need for an explainable and clinically meaningful AI system that not only classifies RBC morphology accurately but also provides interpretable predictions and automated diagnostic reports to support healthcare professionals.

3. Objectives

- Study and analyze the RBC segmentation and classification framework proposed in the selected base paper.
- Develop an EfficientNetV2-based transfer learning model for multiclass RBC morphology classification.
- Improve classification performance using advanced data augmentation and class imbalance handling techniques.
- Generate confidence scores and Top-3 predicted classes for every RBC image.
- Integrate Explainable AI techniques (Grad-CAM) to visualize the regions influencing model predictions.
- Automatically generate a clinical interpretation report describing the predicted morphology and possible disease associations.
- Develop a Streamlit-based web application for real-time RBC morphology analysis.
- Evaluate the proposed system using standard performance metrics and compare it with the baseline approach.

4. Literature Review and Research Gap

Recent studies have demonstrated the effectiveness of deep learning for automated RBC morphology analysis. The selected base paper introduces a comprehensive framework that performs RBC segmentation, overlapping cell separation using ellipse fitting, and multiclass classification while effectively handling severe class imbalance. This approach significantly improves classification performance on blood smear images and addresses challenges associated with overlapping cells.

Despite these contributions, existing methods primarily emphasize classification accuracy and segmentation quality. Most systems do not provide explainability regarding model predictions, making them difficult for clinicians to interpret and trust. Furthermore, automated clinical

report generation and confidence estimation are generally absent, limiting their practical utility in healthcare environments.

Sr. no.	Paper	Contribution	Limitation
1.	Red Blood Cell Segmentation with Overlapping Cell Separation and Classification from an Imbalanced Dataset	Overlapping Cell Separation	Dataset Imbalance Issue
2.	Red Blood Cell Classification Based on Attention Residual Feature Pyramid Network	Attention-Based Classification	Limited Dataset Size
3.	CNN-based Blood Cell Classification	Automatic RBC classification	Image-only prediction
4.	EfficientNet for Blood Smear Analysis	High classification accuracy	Limited explainability
5.	Vision Transformer in Medical Imaging	Better feature extraction	Requires large datasets
6.	Grad-CAM Explainable AI	Visual interpretation	Does not improve prediction
7.	Multi-modal Medical AI	Combines images and clinical data	Limited work on thalassemia

Research Gap: Current RBC morphology classification systems focus mainly on accurate prediction but provide limited support for clinical interpretation. There is a lack of integrated frameworks that combine accurate deep learning-based classification with Explainable AI, prediction confidence, and automated medical reporting. This project addresses these limitations by extending the selected base paper through the integration of EfficientNetV2,

Grad-CAM-based explainability, confidence estimation, and automated clinical report generation within a deployable web application.

5. Proposed Methodology / Plan of Work

The proposed methodology builds upon the framework presented in the selected base paper. Individual RBC images obtained from the segmented dataset undergo preprocessing, normalization, and data augmentation to improve model generalization. An EfficientNetV2-based transfer learning model is employed to extract discriminative morphological features and classify RBC images into nine morphology classes.

To address class imbalance, weighted cross-entropy loss, focal loss, and weighted sampling strategies are incorporated during model training. The trained model predicts the RBC morphology along with confidence scores and Top-3 probable classes.

Grad-CAM is integrated to generate visual explanations by highlighting image regions responsible for the model's prediction. Finally, an automated medical report summarizes the predicted morphology, confidence score, possible clinical significance, and explainability visualization. The complete system is deployed through a Streamlit-based web application for interactive use.

6. Technology, Tools and Platforms

Category	Technology / Tools
Programming Language	Python
Deep Learning Framework	TensorFlow / PyTorch
Deep Learning Models	EfficientNet, Vision Transformer (Optional)
Image Processing	OpenCV, Scikit-image
Machine Learning	Scikit-learn
Explainable AI (XAI)	Grad-CAM, SHAP (Optional)
Dataset	Mendeley RBC Dataset, CBC Parameters Dataset (Hospital/Public Dataset or Simulated Data)
Deployment Platform	Streamlit
Development Environment	Jupyter Notebook / Google Colab / VS Code
Version Control (Optional)	Git & GitHub

8. Project Scope

The proposed system is intended to function as an AI-assisted clinical decision support tool for automated RBC morphology analysis. By extending the selected base paper with Explainable AI and automated clinical reporting, the system enhances transparency and usability for healthcare professionals. It can assist hematologists, pathology laboratories, researchers, and students by providing rapid, consistent, and interpretable RBC classification results.

Future enhancements may include complete blood smear analysis, segmentation of multiple overlapping cells directly from microscope images, disease-level diagnosis through integration with Complete Blood Count (CBC) parameters, cloud deployment, and validation on larger multi-center clinical datasets.

9. Project Timeline

Phase	Duration	Milestone / Deliverable
Literature Review & Base Paper Analysis	Week 1 -2	Study the selected research paper, RBC morphology, existing techniques, and identify research gaps.
Dataset Preparation & Preprocessing	Week 3 - 4	Analyze the Mendeley RBC dataset perform preprocessing, Data Augmentation, and address class imbalance
Model Development & Training	Week 5 - 8	Implement the EfficientNetV2-based classification model and train using transfer learning techniques.
Model Evaluation & Explainable AI	Week 9 - 10	Evaluate the model and integrate Grad-CAM for visual interpretation of predictions.
Clinical Report Generation & Streamlit Deployment	Week 11 -13	Develop the automated clinical report generator and deploy the model.
Testing, Documentation & Final Presentation	Week 14 -16	Perform system testing, prepare documentation, and complete the final project presentation.

10. References

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Group Member Details

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